Template: Innovation Brief

This innovation brief should present an overview of your chosen opportunity. It will help you remember why, what, and how you ended up with the opportunity. It is a crucial tool to bring with you for the second diamond.

1. What

Describe your chosen opportunity:
- What potential value can you create for which stakeholders?

The specific opportunity is to expand local electricity capacity on Bornholm so that BEOF can shift its district heating system from biomass toward electricity.
Bornholm's connection to the wider grid runs through a single subsea cable to Sweden, currently guaranteed at only 35 MW of import capacity, with capacity from 2028 expected to fall short of demand as electrification increases.
Due to end-of-life of the small boiler in Nexø, along with required down scale of biomass use, BEOF plans to shift from a straw boiler to either an electric or hybrid setup.
For BEOF's transition to systems run on electric power, local renewal energy production will need to be expanded, something BEOF's own CEO has flagged as necessary to fully green the system.

The opportunity is therefore to increase the locally available electric power capacity on Bornholm, enabling BEOF to make this transition. Furthermore, this could help the general problem of electric capacity on the island and make them more independent.

Write your Innovation Question from the last activity:
(remember to not describe it as a problem or solution)

How might we secure sufficient electrical capacity in Nexø to enable BEOF to shift its district heating from biomass toward electricity?

2. Why

Argue why your identified opportunity is an opportunity.
- What data (e.g. quotes) from your data research argue for your opportunity?
- Why is there room for value creation?
- Try to be specific: Why do you believe that you can make improvements within your opportunity?

Interviews with the Project Manager of Energy Transition at BEOF confirmed that the current electric power capacity in Nexø is too low to support a shift away from biomass. BEOF itself was already refused permission to install and operate a large electric boiler at its Rønne plant due to grid capacity constraints, and Trefor (the local grid operator) has had to reject several other major electricity projects on the island. The representative from Ocean Prawns, which is a member of a larger forum for local businesses on Bornholm, mentioned that several new local businesses have been refused access the electricity from Sweden and already existing businesses have been prevented from expanding.

The interview with the Project Manager of Innovation at BEOF explained how combining an electric boiler with a renewable energy system and a heat storage system will allow BEOF to use electric power for heat when the price is low and the capacity is sufficient.

This isn't just a benefit for BEOF: expanding capacity here creates room for value beyond the company itself, solving a constraint that has already blocked other Bornholm businesses and supporting the island's broader electrification and renewable-energy ambitions ahead of the Bornholm Energy Island connection in the early 2030s.

3. Where

Where is the opportunity? (A specific company, all universities, nursing homes, etc? Only in Denmark, the EU, developing countries) Be specific and provide reasoning.

This opportunity is mainly located on Bornholm and is especially relevant for BEOF and the local district heating system. Bornholm is currently trying to reduce the use of biomass, but the limited capacity of electricity grid makes it difficult to switch to electricity. Adding local energy production, e.g. a solar plant on the island, could be connected to the grid or directly to a hybrid energy system. The idea could also be used for other islands in the Baltic Sea that have similar district heating systems and limited grid capacity.

4. Who

List stakeholders that are impacted by the opportunity and describe briefly each stakeholder and point out the following characteristics:
- Their relevant behavior connected to the situation in which the opportunity would intervene.
- Their background, wants & needs, and frustrations.
- Relevant areas of interest (e.g. values, focus points).

Remember that stakeholders can be everything from potential users, customers, politicians, companies, organizations...

Stakeholders	Characteristics
BEOF	As BEOF is making the transition from burning biomass to electric boilers, the already insufficient electric energy capacity will become over-burdened. As a producer of electric power, BEOF is also interested in expanding in that area and combing their electric energy and district heat production.
Residents on Bornholm	The residents will be the final users but are not aware of this problem and the influence it has on their lives. If the systems become more independent, the users will have more stable and secure grid, with lower prices. Land on Bornholm would need to be used for the project, which several residents mentioned to oppose.
Local Businesses	Local businesses would not be limited from connecting to the electric cable from Sweden or would not need to as local production would be sufficient.
Ocean Prawns (Local Investors)	Ocean Prawns is one of the largest suppliers of MSC-certified prawns and Greenland halibut. They have an investment portfolio on Bornholm. The businesses have around 250 employees and the portfolio is actively looking for more investment opportunities. During the interview, they expressed interest in investing in sustainable and economically viable projects.
The Municipality of Bornholm	The municipality would need to spend money on a solution, as now local plans would need to be made along with other bureaucratic decisions and work.

5. When

What is the timeframe of the opportunity? When will it arise and be relevant?

This opportunity is relevant now and in the coming years, especially during Bornholm's energy transition towards the introduction of the Baltic Energy Island in 2036. Currently, the electricity grid capacity is limited, while BEOF wants to reduce biomass consumption and use more electricity. Therefore, expanding local electricity energy production is urgent even before the Bornholm Energy Island and new electricity infrastructure become fully operational. A solution should ideally be implemented during the next 5 years.

How frequently does it occur? (every morning on the way to work, multiple times per day, every summer holiday). Is it only for a limited time? (in 2025 a new EU regulation, during the transition to green energy, etc.).

This is a continuous opportunity. It is relevant throughout the energy transition, especially while Bornholm's electricity capacity is limited. It is particularly important in the coming years before the planned offshore wind hub and new electricity infrastructure become fully operational. After that, the importance of this opportunity may decrease or change.

6. How

How big is the opportunity? How many are affected? How much money, impact, time could you save, create or improve?

This opportunity directly impacts the entire Nexø district heating network. By securing local electrical capacity, BEOF can achieve its concrete target of cutting biomass consumption by 50% (aiming for the 30,000 tonne limit by 2050), as stated in the the challenge briefing materials. The current 35 MW limit on the Swedish transmission cable is the primary bottleneck blocking this transition. Solving this capacity shortage prevents the long term economic risk of relying on expensive imported biomass and secures financially viable district heating for the Nexø community.

Which methods did you select on day 6? And how did you execute it? Which of the methods gave you the most insights?

For data collection, we selected active interaction methods. We executed this by conducting semi-structured interviews with officials from the Municipality and the Baltic Energy Island Foundation, alongside active intercept interviews with citizens of Nexø. The active stakeholder interviews gave us the most profound insights. While we initially went in looking for technical answers about district heating mechanics, talking directly to stakeholders revealed that the 35 MW grid bottleneck is the true root of the problem. Our biggest takeaway was that the transition to green district heating in Nexø is physically impossible until this local electric capacity issue is solved.

Template:
Concept Assumptions

Be aware some of your assumptions needs to be tested on day 11 and 12

2. List the most crucial unknown data. (minimum 5).

How much electricity can 8,000 m² of solar panels produce annually?

Potential size of the construction side we could use for our concept.

Current energy and heat storage capacities

Location and geography of the potential building side.

Roof-area solar panels can be built on.

Economic feasibility

Lifetime of the technical components

Would citizens be willing to have solar panels on the houses?

Actual electricity and heat demand of Nexo throughout the year

Efficiency and energy losses of the battery and heat-storage systems.

Can the existing electricity grid support the new energy system?

3. Transform the unknown data into assumptions.

We assume that 8,000 m² of solar panels can produce approximately X MWh of electricity per year.

We assume that the construction site for our concept is at least 15,000 m² large (big enough for the project)

We assume that there is currently 36h of heat storage and 15 MWh of energy storage.

We assume that the construction is close to Nexo and mostly flat terrain.

We assume there is at least 8,000 m² rooftop area for building new solar panels.

We assume the solar energy from solar panels, batteries and the heat system are economically feasible for the community

We assume the main components such as the solar panels, battery etc. will operate for their expected lifetime

We assume that approx. 220 citizens of Nexo are open to the project and do want some positive changes

We assume that our estimated annual and peak electricity and heat demand represents the actual demand of Nexø

We assume that energy losses during storage and distribution are low enough for the system to meet the required demand.

We assume that the existing electricity grid in Nexo has enough capacity to support the solar panels, batteries and increased electricity demand.

4. List the consequences of having a specific wrong assumption.

The system might not meet the demand, especially during winter or less sunny periods

If the available area is smaller than assumed, we may need to reduce the size of the system or find additional locations, which could increase costs and and make the concept less feasible

If there are already more storage capacities we have to build less, but if there are less we have to build more, which would blow the budget and require more space than expected

If the construction side is too far away from Nexo we lose a lot of heat during the transport and would have to reconsider building sides.

Solar production will be too low, and then additional energy sources will be needed

The concept may become too expensive to implement

Replacement costs could make the solution less sustainable and economic

If citizens are unwilling to participate, fewer solar panels can be installed, reducing the total energy production and potentially making the concept unfeasible.

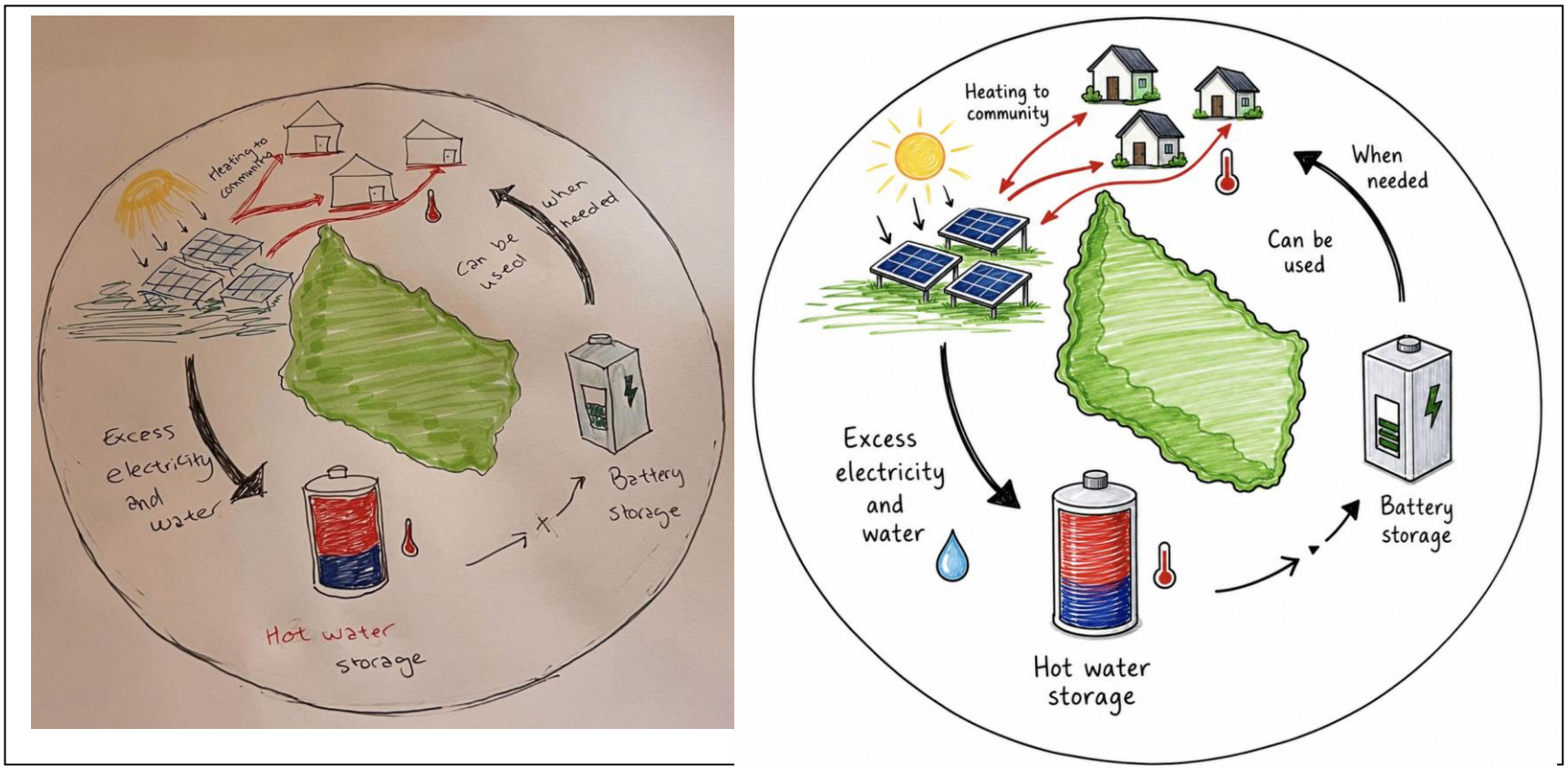
If the demand is higher than expected, the proposed solar panels and storage systems may be undersized and additional energy sources will be required.

If energy losses are higher than expected, more solar panels and storage capacity will be required, increasing both cost and space requirements.

If the grid does not have enough capacity, upgrades may be required, increasing the cost and complexity of the concept.

Template:
Top early-stage concepts

1 Sketch your top early-stage concept in use. Include any useful details.



2 Describe the way your concept functions.

The concept functions as an integrated renewable energy system on Bornholm where solar panels generate electricity to power electric boilers for district heating. Surplus electricity is stored in batteries and excess thermal energy is captured in hot water storage tanks, allowing the system to distribute heat to the community consistently even when solar generation is intermittent, creating a closed loop between generation, storage, and demand.

3 In one sentence, why does this concept matter?

With this concept we are enabling Bornholm's district heating to transition away from biomass by creating a locally generated, renewable energy system that stores excess solar power and thermal energy to maintain reliable heating for the community.